

OBJECT CLASS

Characteristics of Object Class:

1. It is the super-most parent class in Java.
2. All classes in Java implicitly have Is-A relationship with Object class.
3. It is an in-built class, present in java.lang package
4. The fully qualified name is java.lang.Object
5. It has 11 non-static methods

Non-static methods of Object class:

```
C:\Program Files\Java\jdk-21\bin>javap java.lang.Object
Compiled from "Object.java"
public class java.lang.Object {
    public java.lang.Object();
    public final native java.lang.Class<?> getClass();
    public native int hashCode();
    public boolean equals(java.lang.Object);
    protected native java.lang.Object clone() throws java.lang.CloneNotSupportedException;
    public java.lang.String toString();
    public final native void notify();
    public final native void notifyAll();
    public final void wait() throws java.lang.InterruptedException;
    public final void wait(long) throws java.lang.InterruptedException;
    public final void wait(long, int) throws java.lang.InterruptedException;
    protected void finalize() throws java.lang.Throwable;
}
```

Methods in focus:

We are focusing only on four methods of Object class. They are,

1. public String toString()
2. public boolean equals(Object o)
3. public int hashCode()
4. protected void finalize()

The first three methods have default implementation, which we have to over-ride and give new implementation.

public String toString()

Default implementation: The hexadecimal value, that is address is returned when object reference is printed.

New implementation: Instead of returning the address of the object, we over-ride this method to return the attributes (states) of the object.

public boolean equals(Object o)

Default implementation: The address, that is unique and random hexadecimal value, of two objects are compared.

New implementation: Instead of comparing the address of two objects, we over-ride this method to compare the attributes (states) of the two objects.

public int hashCode()

Default implementation: A random and unique integer value for the object is returned.

New implementation: Instead of returning integer value for the object, a random and unique integer value for the attributes (states) is returned.

protected void finalize()

- ❖ It is a method with 'protected' modifier.
- ❖ It pushes the deleted data to the garbage collector.
- ❖ If called again, the data is permanently deleted.
- ❖ It is used for clean-up process of garbage collector and happens internally.

```
25 class Auto
26 {
27     String brand;
28     String color;
29     int wheels;
30     float price;
31     Auto(String brand, String color, int wheels, float price)
32     {
33         this.brand = brand;
34         this.color = color;
35         this.wheels = wheels;
36         this.price = price;
37     }
38     public String toString()
39     {
40         return "brand of auto: "+brand+" color of auto: "+color+" wheels in auto: "+wheels+" price of auto: "+price;
41     }
42     public boolean equals(Object o)
43     {
44         Auto dc = (Auto)o; //DOWNCASTING
45         return this.brand.equals(dc.brand) && this.color.equals(dc.color) && this.wheels==dc.wheels && this.price==dc.price;
46     }
47     public int hashCode()
48     {
49         int hc1 = brand.hashCode();
50         int hc2 = color.hashCode();
51         int hc3 = wheels;
52         int hc4 = (int)price; //float type converted to int
53         return hc1+hc2+hc3+hc4;
54     }
55 }
56 }
57
```

```
1 package javapackage;
2
3 class ObjectClassDriver
4 {
5     public static void main(String[] args)
6     {
7         System.out.println("****TO STRING METHOD****");
8         Auto a = new Auto("piaggio", "yellow", 3, 254627.2f);
9         System.out.println(a);
10        System.out.println("*****EQUALS OBJECT O METHOD*****");
11        Auto a1 = new Auto("bajaj", "green", 3, 324168.2f);
12        System.out.println(a1);
13        Auto a2 = new Auto("bajaj", "green", 3, 324168.2f);
14        System.out.println(a2);
15        System.out.println(a1==a2); //hexadecimal values are not same
16        //NEW IMPLEMENTATION BY EQUALS OBJECT O METHOD
17        System.out.println(a1.equals(a2)); //string values are same
18        System.out.println("*****HASH CODE METHOD*****");
19        System.out.println(a1.hashCode());
20        System.out.println(a2.hashCode());
21        System.out.println(a1.hashCode() == a2.hashCode());
22    }
23 }
24
```

```
<terminated> ObjectClassDriver [Java Application] C:\Users\naray\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64_21.0.6.v2
****TO STRING METHOD****
brand of auto: piaggio color of auto: yellow wheels in auto: 3 price of auto: 254627.2
*****EQUALS OBJECT O METHOD*****
brand of auto: bajaj color of auto: green wheels in auto: 3 price of auto: 324168.2
false
*****HASH CODE METHOD*****
192443074
192443074
true
```

DESIGN TIPS TO ACHIEVE OVER-RIDING OF OBJECT CLASS METHODS:

toString()

- ❖ In a single return statement, we return all attributes using concatenation.

equals(Object o)

- ❖ Mandatory down-casting is required, from Object type to ClassName type.
- ❖ The address of first object is stored in this keyword.
- ❖ The address of second object is stored in down-casted reference variable.
- ❖ Multiple attributes are compared and logical AND operator is used to return a single boolean value.

hashCode()

- ❖ Unique and random integer values are generated for all attributes and their sum is returned as a single integer value.
- ❖ Over-riding equals(Object o) is a pre-condition to over-ride hashCode()
- ❖ hashCode() is over-ridden to cross-check the working of equals(Object o)